International Football Ranking

Iñaki Moreno Estébanez, Iraitz Santafosta Chamorro

***Abstract*—This paper addresses the optimization problem of ranking 25 national soccer teams based on historical match data and additional performance metrics. We formalize the problem using a linear ordering model, where the encoding space is defined as a permutation of the teams, and the search space represents all possible permutations of these teams. The objective function aims to maximize the alignment between the ranking order and a preference matrix derived from match outcomes, incorporating a weighted exponential factor to prioritize higher-ranking teams. To solve the problem, we propose three algorithms: a constructive algorithm, a local search algorithm, and an evolutionary algorithm. These approaches are evaluated and compared in terms of their ability to find optimal rankings that reflect the teams' relative strengths. The results demonstrate the effectiveness of these methods in accurately representing team performance based on historical data and performance metrics, providing insights into the relative strength of national soccer teams.**

***Index Terms*—Soccer ranking, Linear Ordering Problem, optimization, combinatorial algorithms, constructive algorithm, local search, evolutionary algorithm, performance metrics.**

I. Introduction

IN computer science and applied mathematics, optimization problems aim to find the best solution within a set of options under specific constraints and objectives. Given the often large solution spaces involved, efficient methods such as linear programming and heuristic approaches are used. In the context of a Linear Ordering Problem (LOP) for ranking football teams, optimization facilitates structuring the ranking by assessing each team's performance, seeking the most suitable order without evaluating all possible combinations.

II. Description of the problem

In this paper, we will address an optimization problem applied to international soccer. The objective is to find a way to make a ranking taking into account all the international matches of national soccer teams since 1872. Through the use of optimization techniques, we will seek to develop an accurate and efficient system to rank the national teams, taking into account different factors that influence the performance of the teams in international competitions.

This problem can be framed as a Linear Ordering Problem (LOP), a well-known optimization problem in which the goal is to find a linear ordering of a set of elements based on pairwise preferences. In this case, the elements are the national soccer teams, and the preferences are derived from the results of their international matches. The challenge is to process a vast database of match results, including dates, outcomes, and additional parameters, and use this information to establish an accurate ranking of the teams.

The Linear Ordering Problem (LOP) is ideal for this context because it allows us to compare teams pairwise and determine which teams should be ranked higher based on their relative performance. By applying LOP, we will generate a preference matrix, where each entry reflects the relative advantage of one team over another based on their performance in past matches. The goal is to create a ranking system that reflects the teams' performance over time in a fair and precise manner.

III. Data preprocessing

The dataset consists of national team matches from 1872 to 2021, containing information on the date of each match, the two teams involved (home and away), the goals scored by each team, the tournament in which the match was played, and whether the match was held on neutral ground. In addition, for matches that ended in a draw, data from 1967 onward includes records of penalty shootouts.

To gain a better understanding of the dataset, we will examine a graph showing the frequency of matches played between teams. Given the large number of teams in the database, this initial analysis will focus only on the top 25 teams with the highest number of matches played. This approach allows us to identify patterns and relationships among the most active teams and provides a foundation for further analysis of the dataset.

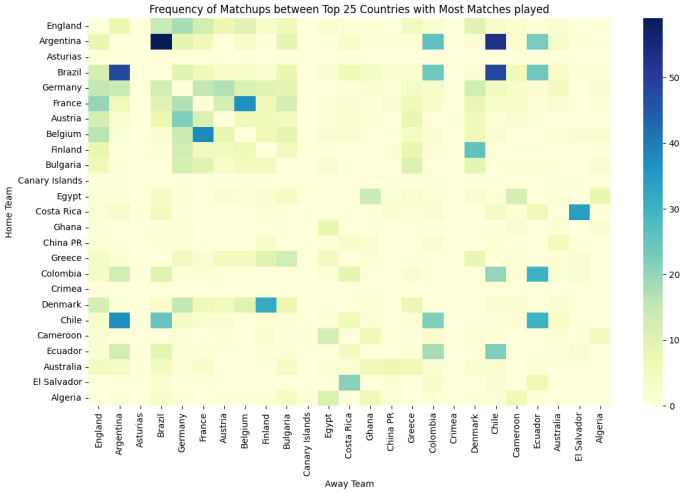

Fig. 1. Illustration of the frequency of matches played between teams

To begin data preprocessing, we examined the database, which contains 311 different national teams and 126 different international tournaments. However, some matches were missing information on either the teams involved or the outcomes, so we decided to exclude these matches from the dataset. Rows containing NaN values in any of the required fields were removed to ensure data consistency.

Given resource constraints, working with all 311 teams to construct the ranking matrix was impractical. Therefore, we focused on the 25 teams with the highest number of matches played in the FIFA World Cup, the tournament we assigned the greatest importance. We also categorized tournaments into five groups, from highest to lowest importance, based on global significance. After applying this filter, the selected teams were: ['Brazil', 'Germany', 'Argentina', 'Italy', 'France', 'England', 'Spain', 'Netherlands', 'Belgium', 'Mexico', 'Sweden', 'Chile', 'Austria', 'South Korea', 'Uruguay', 'Russia', 'Czechoslovakia', 'Hungary', 'Poland', 'Portugal', 'Bulgaria', 'Japan', 'United States', 'Switzerland', 'Paraguay'].

Finally, we defined the parameters to be considered in constructing the matrix. Since relying solely on goals or match outcomes seemed insufficient, we identified additional factors that could enhance ranking accuracy. These parameters are:

Home/Away multiplier: Based on the win, draw, and loss rates for home and away matches, we applied a multiplier to account for home advantage or the challenge of playing away.

Goals multiplier: We calculated the average goals scored per match (home and away) and applied a logarithmic function to reduce the weight of high-scoring matches, providing a more balanced evaluation of each goal.

$$GoalsMulti = (log_2(goals) + 1)/average\ goals \tag{1}$$

Tournament multiplier: We differentiated the value of each tournament based on our classification into five groups, assigning higher multipliers to more prestigious competitions. Draw shootouts: When a match ended in a draw, we checked if it went to a penalty shootout using the "Shootouts.csv" dataset and awarded an additional point to the team that won the shootout.

Data multiplier: We emphasized recent matches over older ones by applying a time-based multiplier, dividing the match year by 2021 (the latest year in the dataset) and adjusting the general match value accordingly.

$$dataMulti = (match\ date)/2021 \tag{2}$$

Applying these parameters, we constructed the LOP matrix by starting with an empty matrix and incrementally calculating the values for each team based on their matches. For each match, we calculated the weighted values for both teams, considering all specified multipliers. These values were then added to their corresponding entries in the matrix, representing the "preference" of one team over another in pairwise comparisons. By the end of this process, we obtained a comprehensive matrix with the 25 selected teams and their aggregated values from all match history, which will serve as the input for the Linear Ordering Problem (LOP) optimization.

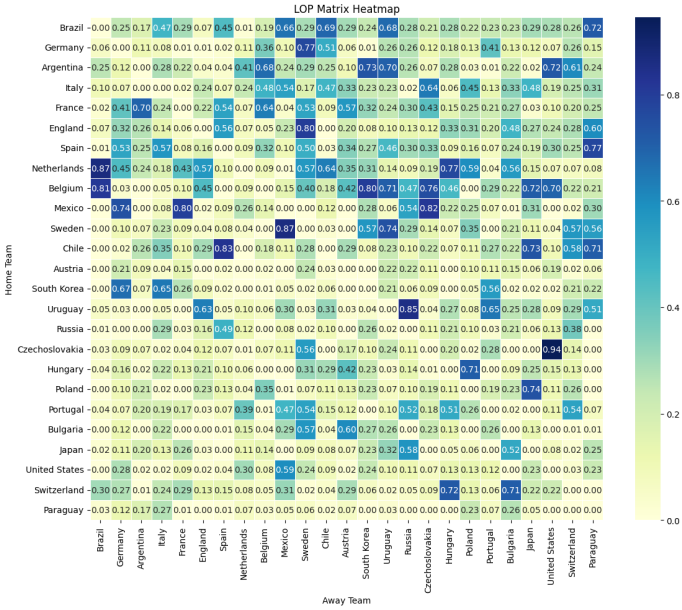

Fig. 2. IFR LOP matrix

This matrix provides a structured representation of team performances, relative importance of matches, and performance context (home, away, neutral ground). With this, the LOP can effectively process the accumulated data to produce a global ranking, identifying the optimal ordering of teams based on their comparative performance in a rigorous and data-driven manner.

IV. Formalizing the Problem

In this study, we aim to formalize the optimization problem of ranking 25 national soccer teams based on historical match data and additional performance metrics.

Encoding Space: the encoding space for this problem is defined as a permutation of a vector representing the 25 selected teams. Each permutation corresponds to a possible order of teams in the ranking, with each position indicating a relative performance score derived from match outcomes and additional evaluative parameters.

Search Space: the search space consists of all possible permutations of the 25 teams (S_n), representing all potential arrangements within the ranking. Since the encoding and search spaces are identical (a bijective representation), every solution within the search space directly corresponds to a

unique permutation in the encoding space. This structure ensures that each permutation is a valid solution and allows for an exhaustive examination of possible team orders to identify the configuration that optimally represents team performance based on historical match data and weighted metrics.

Objective Function: the objective function is designed to maximize alignment between the ranking order and a preference matrix derived from match outcomes and performance metrics. Specifically, this function evaluates each pair of teams within a given permutation and calculates a cumulative score based on pairwise comparisons. To favor teams in higher positions within the ranking, a weighted exponential factor is applied to each position. The mathematical formulation of this objective function can be written as:

$$f(\pi) = \sum_{i=1}^{n} \sum_{j=i+1}^{n} w_i \cdot P(\pi_i, \pi_j) \tag{3}$$

Where:

- $f(\pi)$ is the objective function value for a given permutation π.
- n is the total number of teams.
- i and j represent the indices of the teams in the permutation.
- The weight assigned to the pair of teams based on their ranking position is given by the equation:

$$w_i = (n - i)^{\alpha} \tag{4}$$

  where α controls the emphasis on higher-ranked teams.
- $P(\pi_i, \pi_j)$ is the preference score that quantifies how much the order of teams π_i and π_j aligns with the historical match data.

This objective function evaluates each permutation to find the arrangement that maximizes the overall score, producing a ranking that most accurately reflects each team's relative strength. The exponential weighting applied to higher-ranking teams ensures that top positions are prioritized, adding a stronger emphasis on the leading teams' performance while preserving accurate evaluations across the ranking structure. This method thus identifies the permutation that best aligns with historical performance metrics and provides a comprehensive analysis of team strengths.

V. Algorithm Design

A. Greedy Constructive Algorithm

The constructive algorithm employed in this study is designed to rank national soccer teams based on a performance metric that reflects the overall advantage of each team relative to others. Rather than comparing teams pair by pair, we compute a single performance metric for each team, which is the difference between their total performance and the accumulated performance against them. This metric captures the net strength of a team, considering both their own achievements and how they performed against other teams.

Algorithm Description

1) Metric Calculation: For each team π_i, we calculate a performance metric that represents the difference between the total performance of the team (as indicated by the row sums of the preference matrix) and the total performance accumulated against them (as indicated by the column sums). This metric is computed as follows:

$$\text{Metric}(\pi_i) = \sum_i \text{row}(\pi_i) - \sum_j \text{col}(\pi_i)$$

Where:

- $\sum_i \text{row}(\pi_i)$ represents the performance of team π_i against all other teams.
- $\sum_j \text{col}(\pi_i)$ represents the performance accumulated by team π_i from all other teams.

2) Team Selection: Teams are then ranked according to this performance metric. The team with the highest calculated metric is added first to the ranking, followed by the team with the next highest metric, and so on. This ensures that teams with the greatest net performance advantage are placed at the top of the ranking.

3) Ranking Process: The algorithm proceeds by adding each team sequentially, starting from the team with the highest performance metric, until all teams are included in the ranking. This process systematically builds the ranking from the highest to the lowest-performing teams, based on their aggregate advantage.

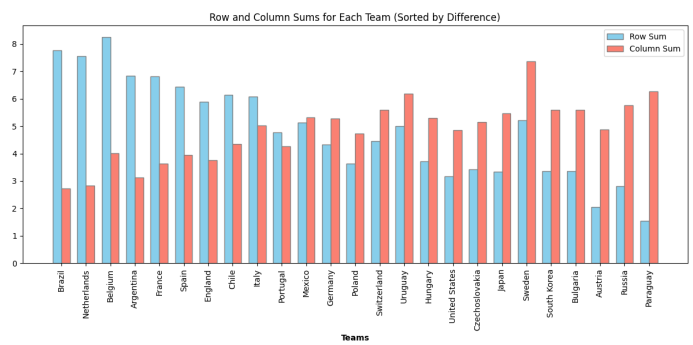

Fig. 3. Difference between each team performance (blue) and against them preformance (red)

The performance-based constructive algorithm provides an efficient means of ranking teams based on their overall performance metrics. Its simplicity allows for fast computation, while still producing a ranking that captures the relative strengths of the teams. This approach offers a practical solution when a balance between solution quality and computational efficiency is necessary, making it well-suited for the problem at hand.

B. Simulated Annealing Algorihtm

The Simulated Annealing (SA) algorithm employed in this study is a probabilistic optimization method used to solve the Linear Ordering Problem (LOP) for ranking national soccer teams. This algorithm is based on the principle of simulating the annealing process, a physical phenomenon where a material cools and solidifies, with the rate of cooling affecting the final structure. In the context of LOP, the algorithm seeks to find a near-optimal ranking of teams by iteratively exploring different team orderings and adjusting the ranking based on an objective function.

Algorithm Description

1) Initialization: The algorithm begins by randomly shuffling the teams to generate an initial solution. The objective function value for this solution is computed, providing a baseline for subsequent iterations. The algorithm also initializes a best_solution variable to store the best ranking found so far.

2) Iterative Optimization: At each iteration, the algorithm generates a neighboring solution by swapping two random teams in the current ranking. This small change simulates a potential improvement in the team's ranking. The objective value for this new arrangement is calculated, and the difference between the new and the previous objective values is evaluated.

3) Acceptance Criterion: If the new solution improves the objective (i.e., the difference is positive), it is always accepted. If the new solution results in a worse objective value, the algorithm may still accept it with a probability determined by the following formula:

$$P = \exp\left(\frac{\Delta f}{T}\right)$$

where Δf is the change in the objective function, and T is the current temperature. This allows the algorithm to escape local optima by accepting some worse solutions early in the process.

4) Cooling Process: After each iteration, the temperature is gradually reduced by a cooling rate. This process lowers the probability of accepting worse solutions over time, focusing the search on refining the solution as the algorithm progresses.

5) Result: The algorithm continues for a specified number of iterations. The solution with the best objective value encountered during this process is returned as the final team ranking.

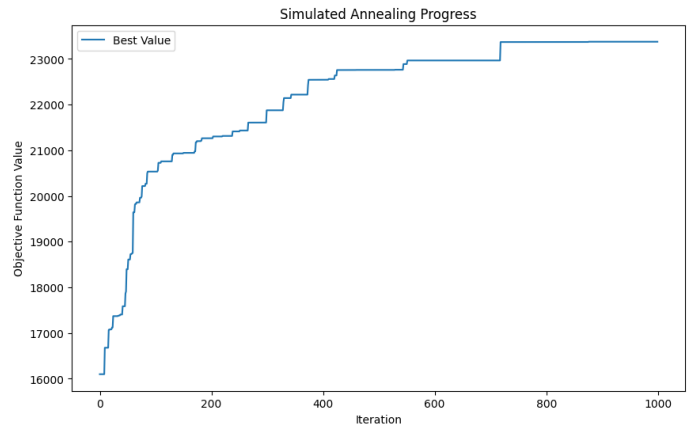

Fig. 4. Simulated Annealing algorithm's progress

The key strength of the Simulated Annealing algorithm lies in its ability to escape local optima and exploring a wide solution space. By probabilistically accepting worse solutions in the early stages and gradually reducing this probability, it balances exploration and exploitation. This makes it particularly well-suited for the Linear Ordering Problem, where exhaustive searches are computationally expensive, offering an effective method for generating high-quality team rankings based on historical match data while maintaining computational efficiency.

C. Improved Estimation of Distribution Algorithm

The Improved Estimation of Distribution Algorithm (EDA) is a probabilistic evolutionary algorithm tailored to solve the Linear Ordering Problem (LOP) for ranking soccer teams based on historical match data. This algorithm utilizes a position-based probability model to guide the search for optimal rankings, enhancing its effectiveness compared to traditional EDAs.

Algorithm Description

1) Initialization: The algorithm starts with a population of randomly generated solutions, each representing a possible ranking of teams.

2) Evolution: This process iterates over several generations, refining the solutions through the following phases:

- Selection: In each generation, the algorithm evaluates the fitness of each solution, which considers match results, goal differences, and tournament importance. The best-performing solutions (those with the highest fitness) are selected to influence the next generation. The selection rate, a parameter of the algorithm, determines the proportion of solutions chosen for this purpose.
- Model Building: Using the selected solutions, the algorithm builds a position-based probabilistic model. This model is represented by a dictionary-like structure, where each key corresponds to a specific position in the ranking. The value associated with each key is another dictionary that stores the probabilities of each team being assigned to that particular position. These probabilities are calculated

based on the frequency with which each team appears in that position within the selected solutions. For example, if 'Brazil' appears in the first position (position 0) in 80% of the selected solutions, the model will assign a probability of 0.8 to 'Brazil' for position 0. This model is built using a specific function for building the position-based model.
- Sampling: The algorithm generates new solutions by sampling from the position-based probability model. For each position in the ranking, a team is selected randomly based on the probabilities defined in the model. This sampling process is implemented using a function that samples from the built model, which handles cases where a team might not have a probability assigned for a specific position due to the randomness in the previous solutions.
- Mutation: To introduce diversity and prevent premature convergence, the algorithm applies a swap mutation to the newly generated solutions with a certain probability (defined by the mutation-rate parameter). The mutation function randomly selects two positions in the ranking and swaps the corresponding teams.
- Replacement: The current population is then replaced with the newly generated and potentially mutated population, ensuring that the algorithm focuses on the most promising solutions in the subsequent generations.

3) Return the Best Solution: After a predetermined number of generations, the algorithm returns the best solution encountered – the ranking of teams that achieves the highest fitness score.

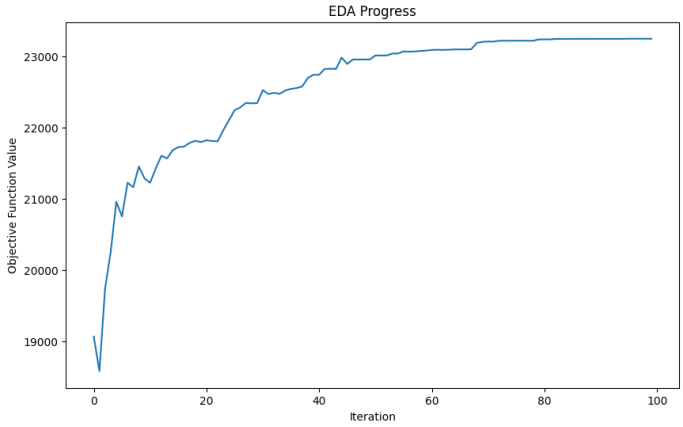

Fig. 5. Simulated Annealing algorithm's progress

The Estimation of Distribution Algorithm (EDA) offers an effective approach for solving the Linear Ordering Problem in the context of soccer team rankings. Its ability to probabilistically learn and adapt the distribution of solutions makes it a powerful tool for generating team rankings based on historical match data, while balancing solution quality and computational efficiency.

VI. Conclusions

In this study, we tackled the challenge of creating an International Football Ranking by formalizing it as a Linear Ordering Problem (LOP). We designed and implemented three distinct algorithms: a constructive Greedy algorithm, a Simulated Annealing algorithm based on local search, and an Estimation of Distribution Algorithm (EDA) representing an evolutionary approach.

A. Evaluation Summary

The following table summarizes the fitness values achieved by each algorithm:

Algorithm	Fitness Value (Maximization)
Greedy	21262.83
Simulated Annealing	23011.54
EDA	23244.77

B. General Conclusions

Based on the evaluation results, we can draw the following conclusions:

1) Performance: The EDA generally outperformed the other two algorithms, achieving the highest fitness value (see Table VI-A). This suggests that the population-based search and probabilistic model building of EDA are more effective in finding high-quality solutions for this specific problem. The Simulated Annealing algorithm also performed well, while the Greedy algorithm obtained a lower fitness value.
2) Computational Cost: While EDA achieved the best results, it's important to consider its computational cost. EDA might require more resources and time compared to the Greedy or Simulated Annealing algorithms. The choice of algorithm depends on the available resources and the desired balance between solution quality and runtime.
3) Solution Similarity: Comparing the solutions, we observed that Greedy Algorithm and Simulated Annealig are the ones with the biggest soulution similarity(with 0.24) and Simulated Annealing with EDA the worst(with 0.079). This could indicate which of them explore similar regions of the search space or coverged toward similar solutions.
4) Overall: Our findings demonstrate the effectiveness of using optimization algorithms for tackling the LOP problem in the context of International Football Ranking. While the EDA showed promising results with the highest fitness value, the choice of algorithm depends on factors such as computational resources and desired solution quality. Further research could explore the impact of different parameter settings, incorporate additional constraints, and compare with other ranking methods.